2106.07890 — formula report

319 inline formulas (section omitted; all rendered), 51 display equations, 0 tables, 6 unrecovered image regions.

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
2106.07890_EQ0004	007	■ 0.255 • C|-6	`\begin{cases}* \rightarrow * & \text { when } d \text { contains } c, \text { red }, \text { and blue } \\ \emptyset \rightarrow * & \text { when } d \text { does not contain both } c \text { and red, and } d \text { does contain blue } \\ \emptyset \rightarrow \emptyset & \text { when } d \text { does not contain both } c \text { and } \mathrm{red}, \text { and } d \text { does not contain blue. }\end{cases}`	$\begin{cases}* \rightarrow * & \text { when } d \text { contains } c, \text { red }, \text { and blue } \\ \emptyset \rightarrow * & \text { when } d \text { does not contain both } c \text { and red, and } d \text { does contain blue } \\ \emptyset \rightarrow \emptyset & \text { when } d \text { does not contain both } c \text { and } \mathrm{red}, \text { and } d \text { does not contain blue. }\end{cases}$	$\begin{cases}* \rightarrow * & \text{when } d \text{ contains } c, \textit{red}, \text{ and } \textit{blue} \\ \emptyset \rightarrow * & \text{when } d \text{ does not contain both } c \text{ and } \textit{red}, \text{ and } d \text{ does contain } \textit{blue} \\ \emptyset \rightarrow \emptyset & \text{when } d \text{ does not contain both } c \text{ and } \textit{red}, \text{ and } d \text{ does not contain } \textit{blue}.\end{cases}$
2106.07890_EQ0012	011	■ 0.403 • W|+1	`\begin{aligned} \mathcal{L}(\text { red,red firetruck }) & =0.02 \\ \mathcal{L}(\text { red,red idea }) & =10^{-5} \\ \mathcal{L}(\text { red, blue sky }) & =0 \end{aligned}`	$\begin{aligned} \mathcal{L}(\text { red,red firetruck }) & =0.02 \\ \mathcal{L}(\text { red,red idea }) & =10^{-5} \\ \mathcal{L}(\text { red, blue sky }) & =0 \end{aligned}$	$\begin{aligned} \mathcal{L}(\textit{red,red firetruck}) &= 0.02 \\ \mathcal{L}(\textit{red,red idea}) &= 10^{-5} \\ \mathcal{L}(\textit{red,blue sky}) &= 0. \end{aligned}$
2106.07890_EQ0030	018	■ 0.508 • N|+1	`\begin{aligned} \mathcal{L}(c, d) \lim _{W} F(c) & = \min \left\{\frac{\mathcal{L}(c, d) f(c)}{w_{1}}, \frac{\mathcal{L}(c, d) g(c)}{w_{2}}, \mathcal{L}(c, d)\right\} \\ & \leq \min \left\{\frac{f(d)}{w_{1}}, \frac{g(d)}{w_{2}}, 1\right\} \\ & \leq \min \left\{\frac{f(d)}{w_{1}}, \frac{g(d)}{w_{2}}, 1\right\} \\ & = \lim _{W} F(d) \end{aligned}`	$\begin{aligned} \mathcal{L}(c, d) \lim _{W} F(c) & = \min \left\{\frac{\mathcal{L}(c, d) f(c)}{w_{1}}, \frac{\mathcal{L}(c, d) g(c)}{w_{2}}, \mathcal{L}(c, d)\right\} \\ & \leq \min \left\{\frac{f(d)}{w_{1}}, \frac{g(d)}{w_{2}}, 1\right\} \\ & \leq \min \left\{\frac{f(d)}{w_{1}}, \frac{g(d)}{w_{2}}, 1\right\} \\ & = \lim _{W} F(d) \end{aligned}$	$\begin{aligned} \mathcal{L}(c, d) \lim_{W} F(c) &= \min \left\{\frac{\mathcal{L}(c, d) f(c)}{w_{1}}, \frac{\mathcal{L}(c, d) g(c)}{w_{2}}, \mathcal{L}(c, d)\right\} \\ &\leq \min \left\{\frac{f(d)}{w_{1}}, \frac{g(d)}{w_{2}}, 1\right\} \\ &\leq \min \left\{\frac{f(d)}{w_{1}}, \frac{g(d)}{w_{2}}, 1\right\} \\ &= \lim_{W} F(d), \end{aligned}$
2106.07890_EQ0024 (16)	017	■ 0.538 • N|+0	`\mathcal{E}\left(\operatorname{colim}_{W} F,-\right) \cong \mathcal{V}^{\mathcal{J}^{\mathrm{op}}}(W, \mathcal{E}(F,-))`	$\mathcal{E}\left(\operatorname{colim}_{W} F,-\right) \cong \mathcal{V}^{\mathcal{J}^{\mathrm{op}}}(W, \mathcal{E}(F,-))$	$\mathcal{E}(\operatorname{colim}_W F, -) \cong \mathcal{V}^{\mathcal{J}^{\mathrm{op}}}(W, \mathcal{E}(F, -))$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>2106.07890_EQ0034</td><td>019</td><td>0.573
W|+0</td><td>\begin{aligned} \widehat{\mathcal{L}}(h, f \times g) & = \inf _{c \in \mathcal{L}}\{[h c,(f \times g) c]\} \\ & = \inf _{c \in \mathcal{L}}\{[h c, \min \{f c, g c\}]\} \\ & =\inf _{c \in \mathcal{L}}\left\{\frac{f c}{h c}, \frac{g c}{h c}, 1\right\} \\ & \left.=\min _{\left\{ \inf _{c \in \mathcal{L}}\right.}\left\{\frac{f c}{h c}, 1\right\}, \inf _{c \in \mathcal{L}}\left\{\frac{g c}{h c}, 1\right\}, 1\right\} \\ & =\widehat{\mathcal{L}}(h, f) \times \widehat{\mathcal{L}}(h, g) \end{aligned}</td><td>$\begin{aligned} \widehat{\mathcal{L}}(h, f \times g) & = \inf _{c \in \mathcal{L}}\{[h c,(f \times g) c]\} \\ & = \inf _{c \in \mathcal{L}}\{[h c, \min \{f c, g c\}]\} \\ & =\inf _{c \in \mathcal{L}}\left\{\frac{f c}{h c}, \frac{g c}{h c}, 1\right\} \\ & =\min _{\{\inf _{c \in \mathcal{L}}}\left\{\frac{f c}{h c}, 1\right\}, \inf _{c \in \mathcal{L}}\left\{\frac{g c}{h c}, 1\right\}, 1\right\} \\ & =\widehat{\mathcal{L}}(h, f) \times \widehat{\mathcal{L}}(h, g) \end{aligned}$</td><td>$\begin{aligned} \widehat{\mathcal{L}}(h, f \times g) & = \inf_{c \in \mathcal{L}}\{[hc,(f \times g)c]\} \\ & = \inf_{c \in \mathcal{L}}\{[hc, \min\{fc, gc\}]\} \\ & = \inf_{c \in \mathcal{L}}\left\{\frac{fc}{hc}, \frac{gc}{hc}, 1\right\} \\ & = \min\left\{\inf_{c \in \mathcal{L}}\left\{\frac{fc}{hc}, 1\right\}, \inf_{c \in \mathcal{L}}\left\{\frac{gc}{hc}, 1\right\}, 1\right\} \\ & = \widehat{\mathcal{L}}(h, f) \times \widehat{\mathcal{L}}(h, g) \end{aligned}$</td></tr>
<tr><td>2106.07890_EQ0044</td><td>021</td><td>0.657
C|+3</td><td>\begin{aligned} {\left[h^{x}, h^{y}\right](c) } & = \widehat{\mathcal{L}}\left(h^{c} \times h^{x}, h^{y}\right) \\ & =\inf _{d \in \mathcal{L}}\left\{\left[\left(h^{c} \times h^{x}\right)(d), h^{y}(d)\right]\right\} \\ & =\inf _{d \in \mathcal{L}}\left\{\frac{\pi(d \mid y)}{\pi(d \mid c) \times \pi(d \mid x)}, 1\right\} \\ & =\inf _{d \in \mathcal{L}}\left\{\frac{\pi(d \mid y)}{\min \{\pi(d \mid c), \pi(d \mid x)\}}, 1\right\} . \end{aligned}</td><td>$\begin{aligned} {\left[h^{x}, h^{y}\right](c) } & = \widehat{\mathcal{L}}\left(h^{c} \times h^{x}, h^{y}\right) \\ & =\inf _{d \in \mathcal{L}}\left\{\left[\left(h^{c} \times h^{x}\right)(d), h^{y}(d)\right]\right\} \\ & =\inf _{d \in \mathcal{L}}\left\{\frac{\pi(d \mid y)}{\pi(d \mid c) \times \pi(d \mid x)}, 1\right\} \\ & =\inf _{d \in \mathcal{L}}\left\{\frac{\pi(d \mid y)}{\min \{\pi(d \mid c), \pi(d \mid x)\}}, 1\right\} . \end{aligned}$</td><td>$\begin{aligned} [h^x, h^y](c) & = \widehat{\mathcal{L}}(h^c \times h^x, h^y) \\ & = \inf_{d \in \mathcal{L}}\{[(h^c \times h^x)(d), h^y(d)]\} \\ & = \inf_{d \in \mathcal{L}}\left\{\frac{\pi(d \mid y)}{\pi(d \mid c) \times \pi(d \mid x)}, 1\right\}. \\ & = \inf_{d \in \mathcal{L}}\left\{\frac{\pi(d \mid y)}{\min\{\pi(d \mid c), \pi(d \mid x)\}}, 1\right\}. \end{aligned}$</td></tr>
<tr><td>2106.07890_EQ0003 (2)</td><td>007</td><td>0.667
N|+0</td><td>\left\{\left(h^{c} \times h^{\text {red }}\right)(d) \rightarrow h^{\text {blue }}(d)\right\}_{d \in \text { ob }(\mathrm{C})}</td><td>$\left\{\left(h^{c} \times h^{\text {red }}\right)(d) \rightarrow h^{\text {blue }}(d)\right\}_{d \in \text { ob }(\mathrm{C})}$</td><td>$\left\{(h^c \times h^{red})(d) \to h^{blue}(d)\right\}_{d \in \mathrm{ob}(\mathsf{C})}$</td></tr>
<tr><td>2106.07890_EQ0020 (12)</td><td>014</td><td>0.776
N|+1</td><td>c \mapsto h^{x}(c):= \begin{cases}\pi(c \mid x) & \text { if } x \leq c \\ 0 & \text { otherwise }\end{cases}</td><td>$c \mapsto h^{x}(c):= \begin{cases}\pi(c \mid x) & \text { if } x \leq c \\ 0 & \text { otherwise }\end{cases}$</td><td>$c \mapsto h^x(c) := \begin{cases}\pi(c \mid x) & \text{if } x \leq c \\ 0 & \text{otherwise.}\end{cases}$</td></tr>
<tr><td>2106.07890_EQ0032 (19)</td><td>018</td><td>0.783
N|-1</td><td>\left(\left(w_{1}, h^{x}\right) \times\left(w_{2}, h^{y}\right)\right)(c)= \begin{cases}\min \left\{\frac{\pi(c \mid x)}{w_{1}}, \frac{\pi(c \mid y)}{w_{2}}, 1\right\} & \text { if } x \leq c \text { and } y \leq c \\ 0 & \text { otherwise }\end{cases}</td><td>$\left(\left(w_{1}, h^{x}\right) \times\left(w_{2}, h^{y}\right)\right)(c)= \begin{cases}\min \left\{\frac{\pi(c \mid x)}{w_{1}}, \frac{\pi(c \mid y)}{w_{2}}, 1\right\} & \text { if } x \leq c \text { and } y \leq c \\ 0 & \text { otherwise }\end{cases}$</td><td>$((w_1, h^x) \times (w_2, h^y))(c) = \begin{cases}\min\left\{\frac{\pi(c \mid x)}{w_1}, \frac{\pi(c \mid y)}{w_2}, 1\right\} & \text{if } x \leq c \text{ and } y \leq c \\ 0 & \text{otherwise.}\end{cases}$</td></tr>
<tr><td>2106.07890_EQ0005</td><td>007</td><td>0.797
C|-4</td><td>\left[h^{x}, h^{y}\right](c)= \begin{cases}* & \text { if every text that contains both } c \text { and } x \text { also contains } y \\ \emptyset & \text { otherwise }\end{cases}</td><td>$\left[h^{x}, h^{y}\right](c)= \begin{cases}* & \text { if every text that contains both } c \text { and } x \text { also contains } y \\ \emptyset & \text { otherwise }\end{cases}$</td><td>$[h^x, h^y](c) = \begin{cases}* & \text{if every text that contains both } c \text{ and } x \text{ also contains } y \\ \emptyset & \text{otherwise.}\end{cases}$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
2106.07890_ EQ0008 (6)	010	■ 0.890 ● N|+0	`[a, b]:= \begin{cases}b / a & \text { if } b<a \\ 1 & \text { otherwise } .\end{cases}`	$[a, b]:= \begin{cases}b / a & \text { if } b<a \\ 1 & \text { otherwise } .\end{cases}$	$[a, b] := \begin{cases} b/a & \textit{if } b < a, \\ 1 & \textit{otherwise}. \end{cases}$
2106.07890_EQ0040	020	■ 0.955 ● N|-1	`\begin{aligned} \mathcal{L}(c, d) \operatorname{colim}_{W} F(c) & =\max \left\{ \mathcal{L}(c, d) w_{1} f(c), \mathcal{L}(c, d) w_{2} g(c)\right\} \\ & \leq \max \left\{\frac{f(d)}{w_{1}}, \frac{g(d)}{w_{2}}\right\} \\ & =\operatorname{colim}_{W} F(d) \end{aligned}`	$\begin{aligned} \mathcal{L}(c, d) \operatorname{colim}_{W} F(c) & =\max \left\{ \mathcal{L}(c, d) w_{1} f(c), \mathcal{L}(c, d) w_{2} g(c)\right\} \\ & \leq \max \left\{\frac{f(d)}{w_{1}}, \frac{g(d)}{w_{2}}\right\} \\ & =\operatorname{colim}_{W} F(d) \end{aligned}$	$\begin{aligned} \mathcal{L}(c, d) \operatorname{colim}_{W} F(c) & =\max \left\{ \mathcal{L}(c, d) w_{1} f(c), \mathcal{L}(c, d) w_{2} g(c)\right\} \\ & \leq \max \left\{\frac{f(d)}{w_{1}}, \frac{g(d)}{w_{2}}\right\} \\ & =\operatorname{colim}_{W} F(d), \end{aligned}$
2106.07890_EQ0048	023	■ 0.967 ● W|+0	`\begin{aligned} \left(\left(w_{1}, f\right) \times \left(w_{2}, g\right)\right)(c) & =-\ln \left(\left(\left(w_{1}^{\prime}, f^{\prime}\right) \times \left(w_{2}^{\prime}, g^{\prime}\right)\right)(c)\right) \\ & =-\ln \left(\min \left\{\frac{f^{\prime} c}{w_{1}^{\prime}}, \frac{g^{\prime} c}{w_{2}^{\prime}}, 1\right \}\right) \\ & =\max \left\{-\ln \left(\frac{f^{\prime} c}{w_{1}^{\prime}}\right),-\ln \left(\frac{g^{\prime} c}{w_{2}^{\prime}}\right), 0\right\} \\ & =\max \left\{- \ln \left(f^{\prime} c\right)+\ln \left(w_{1}^{\prime} \right),-\ln \left(g^{\prime} c\right)+\ln \left(w_{2}^{\prime}\right), 0\right\} \\ & =\max \left\{f c-w_{1}, g c-w_{2}, 0\right\} \end{aligned}`	$\begin{aligned} \left(\left(w_{1}, f\right) \times \left(w_{2}, g\right)\right)(c) & =-\ln \left(\left(\left(w_{1}^{\prime}, f^{\prime}\right) \times \left(w_{2}^{\prime}, g^{\prime}\right)\right)(c)\right) \\ & =-\ln \left(\min \left\{\frac{f^{\prime} c}{w_{1}^{\prime}}, \frac{g^{\prime} c}{w_{2}^{\prime}}, 1\right \}\right) \\ & =\max \left\{-\ln \left(\frac{f^{\prime} c}{w_{1}^{\prime}}\right),-\ln \left(\frac{g^{\prime} c}{w_{2}^{\prime}}\right), 0\right\} \\ & =\max \left\{- \ln \left(f^{\prime} c\right)+\ln \left(w_{1}^{\prime} \right),-\ln \left(g^{\prime} c\right)+\ln \left(w_{2}^{\prime}\right), 0\right\} \\ & =\max \left\{f c-w_{1}, g c-w_{2}, 0\right\} \end{aligned}$	$\begin{aligned} ((w_1, f) \times (w_2, g))(c) & = -\ln\left(((w_1', f') \times (w_2', g'))(c)\right) \\ & = -\ln\left(\min\left\{\frac{f'c}{w_1'}, \frac{g'c}{w_2'}, 1\right\}\right) \\ & = \max\left\{-\ln\left(\frac{f'c}{w_1'}\right), -\ln\left(\frac{g'c}{w_2'}\right), 0\right\} \\ & = \max\left\{-\ln(f'c) + \ln(w_1'), -\ln(g'c) + \ln(w_2'), 0\right\} \\ & = \max\left\{fc - w_1, gc - w_2, 0\right\}. \end{aligned}$
2106.07890_EQ0029	017	■ 0.990 ● W|+0	`\begin{aligned} {[0,1]^{\mathcal{J}}(W, \widehat{ \mathcal{L}}(Z, F)) } & =\min \left\{\left[w_{1}, \inf _{c \in \mathcal{L}}\left\{\frac{f c}{Z c}, 1\right\} \right],\left[w_{2}, \inf _{c \in \mathcal{L}}\left\{ \frac{g c}{Z c}, 1\right\}\right]\right\} \\ & =\inf _{c \in \mathcal{L}}\left\{\frac{f c}{w_{1} Z c}, \frac{g c}{w_{2} Z c}, 1\right\} . \end{aligned}`	$\begin{aligned} {[0,1]^{\mathcal{J}}(W, \widehat{ \mathcal{L}}(Z, F)) } & =\min \left\{\left[w_{1}, \inf _{c \in \mathcal{L}}\left\{\frac{f c}{Z c}, 1\right\} \right],\left[w_{2}, \inf _{c \in \mathcal{L}}\left\{ \frac{g c}{Z c}, 1\right\}\right]\right\} \\ & =\inf _{c \in \mathcal{L}}\left\{\frac{f c}{w_{1} Z c}, \frac{g c}{w_{2} Z c}, 1\right\} . \end{aligned}$	$\begin{aligned} [0,1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(Z, F)) & = \min\left\{\left[w_1, \inf_{c \in \mathcal{L}}\left\{\frac{fc}{Zc}, 1\right\}\right], \left[w_2, \inf_{c \in \mathcal{L}}\left\{\frac{gc}{Zc}, 1\right\}\right]\right\} \\ & = \inf_{c \in \mathcal{L}}\left\{\frac{fc}{w_1 Zc}, \frac{gc}{w_2 Zc}, 1\right\}. \end{aligned}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
2106.07890_EQ0037	019	■ 0.996 ● W|-2	`\begin{aligned} \widehat{\mathcal{L}}\left(\operatorname{colim}_{W} F i, Z\right) & =\inf _{c \in \mathcal{L}}\left\{\left[\max \left\{w_{1} f c, w_{2} g c\right\}, Z c\right]\right\} \\ & =\inf _{c \in \mathcal{L}}\left\{\frac{Z c}{\max \left\{w_{1} f c, w_{2} g c\right\}}, 1\right\} \\ & =\inf _{c \in \mathcal{L}}\left\{\frac{Z c}{w_{1} f c}, \frac{Z c}{w_{2} g c}\right\} . \end{aligned}`	$\begin{aligned} \widehat{\mathcal{L}}\left(\operatorname{colim}_{W} F i, Z\right) & =\inf _{c \in \mathcal{L}}\left\{\left[\max \left\{w_{1} f c, w_{2} g c\right\}, Z c\right]\right\} \\ & =\inf _{c \in \mathcal{L}}\left\{\frac{Z c}{\max \left\{w_{1} f c, w_{2} g c\right\}}, 1\right\} \\ & =\inf _{c \in \mathcal{L}}\left\{\frac{Z c}{w_{1} f c}, \frac{Z c}{w_{2} g c}\right\} . \end{aligned}$	$\begin{aligned} \widehat{\mathcal{L}}(\operatorname{colim}_{W} F i, Z) & =\inf _{c \in \mathcal{L}}\{[\max \{w_{1} f c, w_{2} g c\}, Z c]\} \\ & =\inf _{c \in \mathcal{L}}\left\{\frac{Z c}{\max \{w_{1} f c, w_{2} g c\}}, 1\right\} \\ & =\inf _{c \in \mathcal{L}}\left\{\frac{Z c}{w_{1} f c}, \frac{Z c}{w_{2} g c}\right\} . \end{aligned}$
2106.07890_EQ0039	020	■ 0.997 ● W|+0	`\begin{aligned} {[0,1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(F, Z)) } & =\min \left\{\left[w_{1}, \inf _{c \in \mathcal{L}}\left\{\frac{Z c}{f c}, 1\right\}\right],\left[w_{2}, \inf _{c \in \mathcal{L}}\left\{\frac{Z c}{g c}, 1\right\}\right]\right\} \\ & =\inf _{c \in \mathcal{L}}\left\{\frac{Z c}{w_{1} f c}, \frac{Z c}{w_{2} g c}, 1\right\} . \end{aligned}`	$\begin{aligned} {[0,1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(F, Z)) } & =\min \left\{\left[w_{1}, \inf _{c \in \mathcal{L}}\left\{\frac{Z c}{f c}, 1\right\}\right],\left[w_{2}, \inf _{c \in \mathcal{L}}\left\{\frac{Z c}{g c}, 1\right\}\right]\right\} \\ & =\inf _{c \in \mathcal{L}}\left\{\frac{Z c}{w_{1} f c}, \frac{Z c}{w_{2} g c}, 1\right\} . \end{aligned}$	$\begin{aligned} [0,1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(F, Z)) & =\min \left\{\left[w_{1}, \inf _{c \in \mathcal{L}}\left\{\frac{Z c}{f c}, 1\right\}\right],\left[w_{2}, \inf _{c \in \mathcal{L}}\left\{\frac{Z c}{g c}, 1\right\}\right]\right\} \\ & =\inf _{c \in \mathcal{L}}\left\{\frac{Z c}{w_{1} f c}, \frac{Z c}{w_{2} g c}, 1\right\} . \end{aligned}$
2106.07890_EQ0035	019	■ 0.999 ● N|+0	`\left(w_{1}, f\right) \sqcup\left(w_{2}, g\right):= \operatorname{colim}_{W} F .`	$\left(w_{1}, f\right) \sqcup\left(w_{2}, g\right):= \operatorname{colim}_{W} F .$	$(w_{1}, f) \sqcup (w_{2}, g):= \operatorname{colim}_{W} F .$
2106.07890_EQ0001	007	■ 1.000 ● N|+0	`\left[h^{\text {red }}, h^{\text {blue }}\right]: \mathrm{C} \rightarrow \text { Set }`	$\left[h^{\text {red }}, h^{\text {blue }}\right]: \mathrm{C} \rightarrow \text { Set }$	$[h^{red}, h^{blue}]: \mathsf{C} \rightarrow \mathsf{Set}$
2106.07890_EQ0010	010	■ 1.000 ● N|+0	`\begin{aligned} & a \times b:=\min \{a, b\} \\ & a \sqcup b:=\max \{a, b\} . \end{aligned}`	$\begin{aligned} & a \times b:=\min \{a, b\} \\ & a \sqcup b:=\max \{a, b\} . \end{aligned}$	$\begin{aligned} & a \times b:=\min \{a, b\} \\ & a \sqcup b:=\max \{a, b\} . \end{aligned}$
2106.07890_EQ0002 (1)	007	■ 1.000 ● N|+1	`\left[h^{\text {red }}, h^{\text {blue }}\right](c)= \widehat{\mathrm{C}}\left(h^{c},\left[h^{\text {red }}, h^{\text {blue }}\right]\right)=\widehat{\mathrm{C}} \left(h^{c} \times h^{\text {red }}, h^{\text {blue }} \right) .`	$\left[h^{\text {red }}, h^{\text {blue }}\right](c)= \widehat{\mathrm{C}}\left(h^{c},\left[h^{\text {red }}, h^{\text {blue }}\right]\right)=\widehat{\mathrm{C}} \left(h^{c} \times h^{\text {red }}, h^{\text {blue }} \right) .$	$[h^{red}, h^{blue}](c)= \widehat{\mathsf{C}}(h^{c},[h^{red}, h^{blue}])=\widehat{\mathsf{C}}(h^{c} \times h^{red}, h^{blue}) .$
2106.07890_EQ0006 (4)	009	■ 1.000 ● N|+0	`\begin{gathered} 1 \leq \mathcal{C}(x, x) \\ \mathcal{C}(y, z) \otimes \mathcal{C}(x, y) \leq \mathcal{C}(x, z) \end{gathered}`	$\begin{gathered} 1 \leq \mathcal{C}(x, x) \\ \mathcal{C}(y, z) \otimes \mathcal{C}(x, y) \leq \mathcal{C}(x, z) \end{gathered}$	$\begin{gathered} 1 \leq \mathcal{C}(x, x) \\ \mathcal{C}(y, z) \otimes \mathcal{C}(x, y) \leq \mathcal{C}(x, z) \end{gathered}$
2106.07890_EQ0007 (5)	009	■ 1.000 ● N|-1	`x \otimes y \leq z \text { if and only if } x \leq[y, z]`	$x \otimes y \leq z \text { if and only if } x \leq[y, z]$	$x \otimes y \leq z \text { if and only if } x \leq[y, z]$
2106.07890_EQ0009 (7)	010	■ 1.000 ● N|-1	`a b \leq c \text { if and only if } a \leq[b, c]`	$a b \leq c \text { if and only if } a \leq[b, c]$	$a b \leq c \text { if and only if } a \leq[b, c]$
2106.07890_EQ0011	011	■ 1.000 ● N|+0	`\mathcal{L}(x, y):=\pi(y \mid x),`	$\mathcal{L}(x, y):=\pi(y \mid x),$	$\mathcal{L}(x, y):=\pi(y\|x),$
2106.07890_EQ0013 (8)	011	■ 1.000 ● N|+0	`\pi(z \mid y) \pi(y \mid x)=\pi(z \mid x)`	$\pi(z \mid y) \pi(y \mid x)=\pi(z \mid x)$	$\pi(z\|y)\pi(y\|x)=\pi(z\|x)$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
2106.07890_ EQ0014 (9)	012	■1.000 ● N|+0	`\mathcal{C}(x, y) \leq \mathcal{D}(f x, f y)`	$\mathcal{C}(x, y) \leq \mathcal{D}(f x, f y)$	$\mathcal{C}(x, y) \leq \mathcal{D}(f x, f y)$
2106.07890_ EQ0015 (10)	012	■1.000 ● N|+0	`\mathcal{D}^{\mathcal{C}}(f, g):=\int_{c \in \mathcal{C}} \mathcal{D}(f c, g c)`	$\mathcal{D}^{\mathcal{C}}(f, g):=\int_{c \in \mathcal{C}} \mathcal{D}(f c, g c)$	$\mathcal{D}^{\mathcal{C}}(f, g):=\int_{c \in \mathcal{C}} \mathcal{D}(f c, g c)$
2106.07890_ EQ0016 (11)	013	■1.000 ● N|+0	`\widehat{\mathcal{C}}(f, g)=\inf _{c \in \mathcal{C}}\{[f c, g c]\}`	$\widehat{\mathcal{C}}(f, g)=\inf _{c \in \mathcal{C}}\{[f c, g c]\}$	$\widehat{\mathcal{C}}(f, g)=\inf _{c \in \mathcal{C}}\{[f c, g c]\}$
2106.07890_EQ0017	013	■1.000 ● N|+0	`h^{x}:=\mathcal{C}(x,-)`	$h^{x}:=\mathcal{C}(x,-)$	$h^{x}:=\mathcal{C}(x,-)$
2106.07890_EQ0018	013	■1.000 ● N|+0	`\mathcal{C}(c, d) \leq\left[h^{x}(c), h^{x}(d)\right]`	$\mathcal{C}(c, d) \leq\left[h^{x}(c), h^{x}(d)\right]$	$\mathcal{C}(c, d) \leq\left[h^{x}(c), h^{x}(d)\right]$
2106.07890_EQ0019	013	■1.000 ● N|+0	`\widehat{\mathcal{C}}\left(h^{x}, f\right) \leq \left[h^{x}(x), f x\right]=[1, f x]=f x .`	$\widehat{\mathcal{C}}\left(h^{x}, f\right) \leq \left[h^{x}(x), f x\right]=[1, f x]=f x .$	$\widehat{\mathcal{C}}\left(h^{x}, f\right) \leq \left[h^{x}(x), f x\right]=[1, f x]=f x .$
2106.07890_EQ0021	015	■1.000 ● N|+0	`x \xrightarrow{a} y \quad h^{y} \xrightarrow{a} h^{x}`	$x \xrightarrow{a} y \quad h^{y} \xrightarrow{a} h^{x}$	$x \xrightarrow{a} y \qquad h^{y} \xrightarrow{a} h^{x}$
2106.07890_ EQ0022 (13)	016	■1.000 ● N|+0	`\mathrm{C}(-, \lim F) \cong \operatorname{Set}^{ \mathrm{J}}(*, \mathrm{C}(-, F))`	$\mathrm{C}(-, \lim F) \cong \operatorname{Set}^{ \mathrm{J}}(*, \mathrm{C}(-, F))$	$\mathrm{C}(-, \lim F) \cong \operatorname{Set}^{ \mathrm{J}}(*, \mathrm{C}(-, F))$
2106.07890_ EQ0023 (15)	016	■1.000 ● N|+0	`\mathcal{E}\left(-, \lim _{W} F\right) \cong \mathcal{V}^{\mathcal{J}}(W, \mathcal{E}(-, F))`	$\mathcal{E}\left(-, \lim _{W} F\right) \cong \mathcal{V}^{\mathcal{J}}(W, \mathcal{E}(-, F))$	$\mathcal{E}(-, \lim _{W} F) \cong \mathcal{V}^{\mathcal{J}}(W, \mathcal{E}(-, F))$
2106.07890_EQ0025	017	■1.000 ● K|+0	`\left(w_{1}, f\right) \times\left(w_{2}, g\right):=\lim _{W} F .`	$\left(w_{1}, f\right) \times\left(w_{2}, g\right):=\lim _{W} F .$	$\left(w_{1}, f\right) \times\left(w_{2}, g\right):=\lim _{W} F .$
2106.07890_ EQ0026 (17)	017	■1.000 ● N|+0	`\widehat{\mathcal{L}}\left(Z, \lim _{W} F\right) \cong[0,1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(Z, F)) .`	$\widehat{\mathcal{L}}\left(Z, \lim _{W} F\right) \cong[0,1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(Z, F)) .$	$\widehat{\mathcal{L}}(Z, \lim _{W} F) \cong[0,1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(Z, F)) .$
2106.07890_EQ0027	017	■1.000 ● W|+0	`\inf _{c \in \mathcal{L}}\left\{\left[Z c, \min \left\{\frac{f c}{w_{1}}, \frac{g c}{w_{2}}, 1\right\}\right]\right\}=\inf _{c \in \mathcal{L}}\left\{\frac{f c}{w_{1} Z c}, \frac{g c}{w_{2} Z c}, 1\right\} .`	$\inf _{c \in \mathcal{L}}\left\{\left[Z c, \min \left\{\frac{f c}{w_{1}}, \frac{g c}{w_{2}}, 1\right\}\right]\right\}=\inf _{c \in \mathcal{L}}\left\{\frac{f c}{w_{1} Z c}, \frac{g c}{w_{2} Z c}, 1\right\} .$	$\inf _{c \in \mathcal{L}}\left\{\left[Z c, \min \left\{\frac{f c}{w_{1}}, \frac{g c}{w_{2}}, 1\right\}\right]\right\}=\inf _{c \in \mathcal{L}}\left\{\frac{f c}{w_{1} Z c}, \frac{g c}{w_{2} Z c}, 1\right\} .$
2106.07890_EQ0028	017	■1.000 ● W|+0	`\widehat{\mathcal{L}}(Z, F i)=\inf _{c \in \mathcal{L}} \{[Z c, F i c]\}=\inf _{c \in \mathcal{L}}\left\{\frac{F i c}{Z c}, 1\right\} .`	$\widehat{\mathcal{L}}(Z, F i)=\inf _{c \in \mathcal{L}} \{[Z c, F i c]\}=\inf _{c \in \mathcal{L}}\left\{\frac{F i c}{Z c}, 1\right\} .$	$\widehat{\mathcal{L}}(Z, F i)=\inf _{c \in \mathcal{L}} \{[Z c, F i c]\}=\inf _{c \in \mathcal{L}}\left\{\frac{F i c}{Z c}, 1\right\} .$
2106.07890_ EQ0031 (18)	018	■1.000 ● N|+0	`\left(\left(w_{1}, h^{x}\right) \times\left(w_{2}, h^{y} \right)\right)(c)=\min \left\{\frac{h^{x}(c)}{w_{1}}, \frac{h^{y}(c)}{w_{2}}, 1\right\} .`	$\left(\left(w_{1}, h^{x}\right) \times\left(w_{2}, h^{y} \right)\right)(c)=\min \left\{\frac{h^{x}(c)}{w_{1}}, \frac{h^{y}(c)}{w_{2}}, 1\right\} .$	$((w_{1}, h^{x}) \times(w_{2}, h^{y}))(c)=\min \left\{\frac{h^{x}(c)}{w_{1}}, \frac{h^{y}(c)}{w_{2}}, 1\right\} .$
2106.07890_EQ0033	018	■1.000 ● N|+0	`\widehat{\mathcal{L}}(h, f \times g)=\widehat{ \mathcal{L}}(h, f) \times \widehat{\mathcal{L}}(h, g) .`	$\widehat{\mathcal{L}}(h, f \times g)=\widehat{ \mathcal{L}}(h, f) \times \widehat{\mathcal{L}}(h, g) .$	$\widehat{\mathcal{L}}(h, f \times g)=\widehat{ \mathcal{L}}(h, f) \times \widehat{\mathcal{L}}(h, g) .$
2106.07890_ EQ0036 (20)	019	■1.000 ● K|+0	`\widehat{\mathcal{L}}\left(\operatorname{colim}_{W} F, Z \right)=[0,1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(F, Z)) .`	$\widehat{\mathcal{L}}\left(\operatorname{colim}_{W} F, Z \right)=[0,1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(F, Z)) .$	$\widehat{\mathcal{L}}(\operatorname{colim}_{W} F, Z)=[0,1]^{\mathcal{J}}(W, \widehat{\mathcal{L}}(F, Z)) .$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

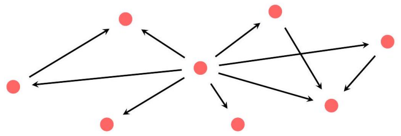
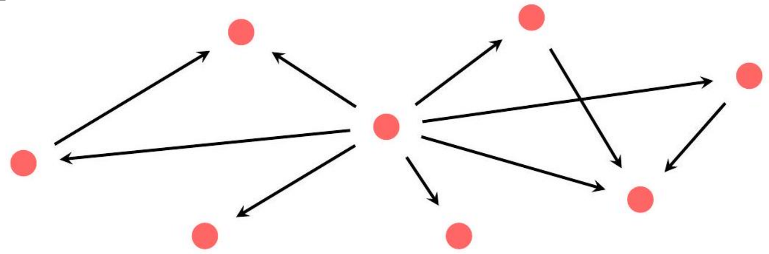
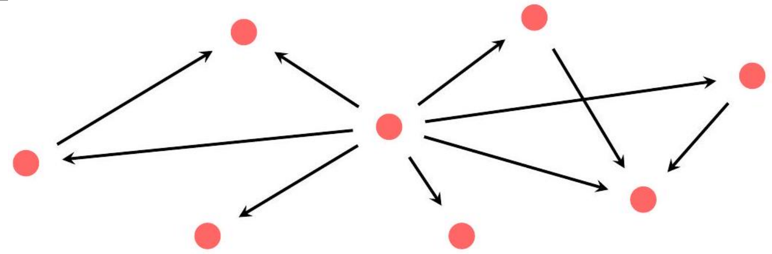
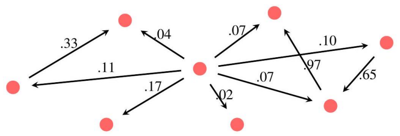
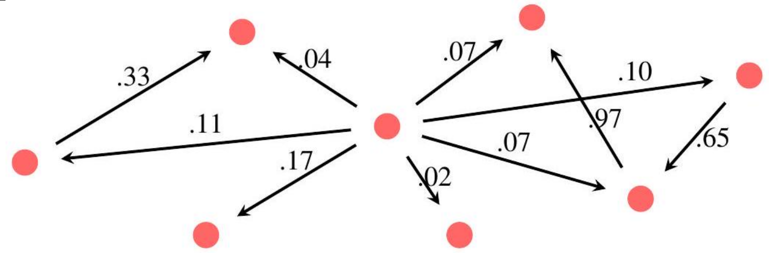
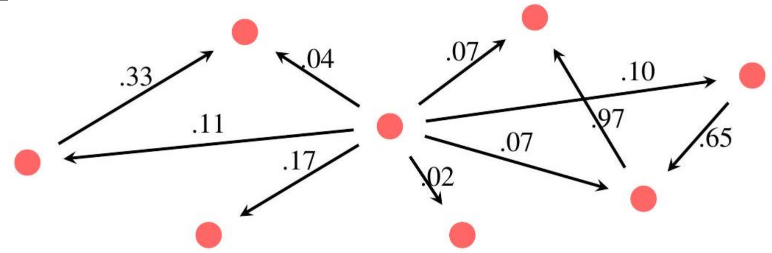
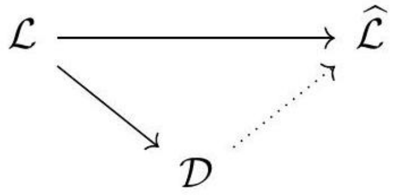
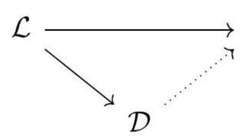
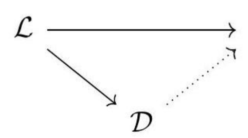
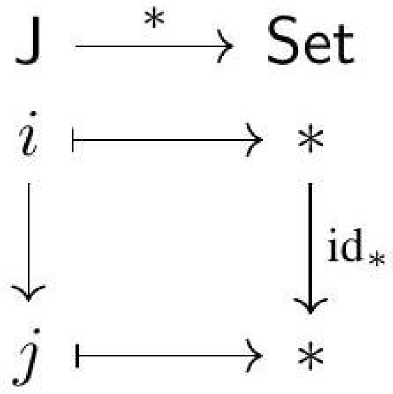
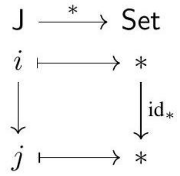
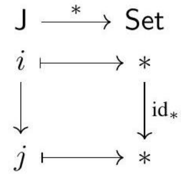
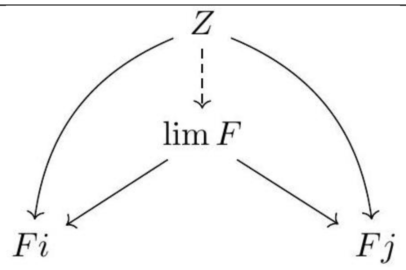
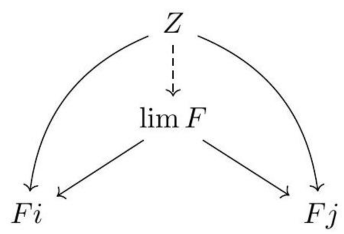
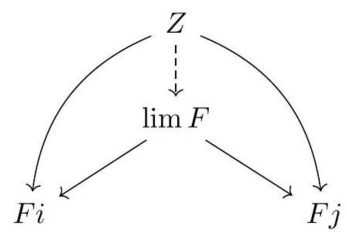
Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
2106.07890_EQ0038	019	■ 1.000 ● W|+0	`\widehat{\mathcal{L}}(F i, Z)=\inf _{c \in \mathcal{L}} \{[F i c, Z c]\}=\inf _{c \in \mathcal{L}}\left\{\frac{Z c}{F i c}, 1\right\} .`	$\widehat{\mathcal{L}}(F i, Z)=\inf _{c \in \mathcal{L}} \{[F i c, Z c]\}=\inf _{c \in \mathcal{L}}\left\{\frac{Z c}{F i c}, 1\right\} .$	$\widehat{\mathcal{L}}(F i, Z)=\inf _{c \in \mathcal{L}} \{[F i c, Z c]\}=\inf _{c \in \mathcal{L}}\left\{\frac{Z c}{F i c}, 1\right\} .$
2106.07890_EQ0041	020	■ 1.000 ● N|+2	`[f, g](c):=\widehat{\mathcal{L}}\left(h^{c} \times f, g \right) .`	$[f, g](c):=\widehat{\mathcal{L}}\left(h^{c} \times f, g \right) .$	$[f, g](c):=\widehat{\mathcal{L}}(h^{c} \times f, g).$
2106.07890_EQ0042	021	■ 1.000 ● N|+0	`\widehat{\mathcal{L}}(h \times f, g)=\widehat{ \mathcal{L}}(h,[f, g]) .`	$\widehat{\mathcal{L}}(h \times f, g)=\widehat{\mathcal{L}}(h,[f, g]).$	$\widehat{\mathcal{L}}(h \times f, g)=\widehat{\mathcal{L}}(h,[f, g]).$
2106.07890_ EQ0043 (21)	021	■ 1.000 ● N|+0	`\left[h^{x}, h^{y}\right](c)=\inf _{d \in \mathcal{L}} \left\{\frac{\pi(d \mid y)}{\min \{\pi(d \mid c), \pi(d \mid x)\}}, 1\right\} .`	$\left[h^{x}, h^{y}\right](c)=\inf _{d \in \mathcal{L}} \left\{\frac{\pi(d \mid y)}{\min \{\pi(d \mid c), \pi(d \mid x)\}}, 1\right\} .$	$[h^{x}, h^{y}](c)=\inf _{d \in \mathcal{L}} \left\{\frac{\pi(d \mid y)}{\min \{\pi(d \mid c), \pi(d \mid x)\}}, 1\right\} .$
2106.07890_EQ0045	023	■ 1.000 ● N|+0	`\widehat{\mathcal{M}}(f, g)=\sup _{x \in \mathcal{M}} d_{[0, \infty]}(f x, g x)=\sup _{x \in \mathcal{M}} \max \{g x-f x, 0\}`	$\widehat{\mathcal{M}}(f, g)=\sup _{x \in \mathcal{M}} d_{[0, \infty]}(f x, g x)=\sup _{x \in \mathcal{M}} \max \{g x-f x, 0\}$	$\widehat{\mathcal{M}}(f, g)=\sup _{x \in \mathcal{M}} d_{[0, \infty]}(f x, g x)=\sup _{x \in \mathcal{M}} \max \{g x-f x, 0\}$
2106.07890_EQ0046	023	■ 1.000 ● W|+0	`\begin{gathered} \left(\left(w_{1}, f\right) \times \left(w_{2}, g\right)\right)(c)=\max \left\{f c-w_{1}, g c-w_{2}, 0\right\} \\ \left(\left(w_{1}, f\right) \sqcup \left(w_{2}, g\right)\right)(c)=\min \left\{f c+w_{1}, g c+w_{2}\right\} \end{gathered}`	$\begin{gathered} \left(\left(w_{1}, f\right) \times \left(w_{2}, g\right)\right)(c)=\max \left\{f c-w_{1}, g c-w_{2}, 0\right\} \\ \left(\left(w_{1}, f\right) \sqcup \left(w_{2}, g\right)\right)(c)=\min \left\{f c+w_{1}, g c+w_{2}\right\} \end{gathered}$	$\begin{gathered} ((w_{1}, f) \times (w_{2}, g))(c)=\max \{f c-w_{1}, g c-w_{2}, 0\} \\ ((w_{1}, f) \sqcup (w_{2}, g))(c)=\min \{f c+w_{1}, g c+w_{2}\} \end{gathered}$
2106.07890_EQ0047	023	■ 1.000 ● N|+0	`\left(\left(w_{1}^{\prime}, f^{\prime}\right) \times \left(w_{2}^{\prime}, g^{\prime}\right)\right)(c)=\min \left\{\frac{f^{\prime} c}{w_{1}^{\prime}}, \frac{g^{ \prime} c}{w_{2}^{\prime}}, 1\right\}`	$\left(\left(w_{1}^{\prime}, f^{\prime}\right) \times \left(w_{2}^{\prime}, g^{\prime}\right)\right)(c)=\min \left\{\frac{f^{\prime} c}{w_{1}^{\prime}}, \frac{g^{\prime} c}{w_{2}^{\prime}}, 1\right\}$	$((w_{1}^{\prime}, f^{\prime}) \times (w_{2}^{\prime}, g^{\prime}))(c)=\min \left\{\frac{f^{\prime} c}{w_{1}^{\prime}}, \frac{g^{\prime} c}{w_{2}^{\prime}}, 1\right\}$
2106.07890_EQ0049	024	■ 1.000 ● W|+1	`\begin{aligned} \left(\left(w_{1}, f\right) \sqcup \left(w_{2}, g\right)\right)(c) & =-\ln \left(\left(\left(w_{1}^{\prime}, f^{\prime}\right) \sqcup \left(w_{2}^{\prime}, g^{\prime}\right)\right)(c)\right) \\ & =-\ln \left(\max \left\{w_{1}^{\prime} f^{\prime} c, w^{\prime} 2 g^{\prime} c\right\}\right) \\ & =\min \left\{-\ln \left(w_{1}^{\prime} f^{\prime} c\right),-\ln \left(w_{2}^{\prime} g^{\prime} c\right)\right\} \\ & =\min \left\{-\ln \left(f^{\prime} c\right)-\ln \left(w_{1}^{\prime}\right),-\ln \left(g^{\prime} c \right)-\ln \left(w_{2}^{\prime}\right)\right\} \\ & = \min \left\{f c+w_{1}, g c+w_{2}\right\} . \end{aligned}`	$\begin{aligned} \left(\left(w_{1}, f\right) \sqcup \left(w_{2}, g\right)\right)(c) & =-\ln \left(\left(\left(w_{1}^{\prime}, f^{\prime}\right) \sqcup \left(w_{2}^{\prime}, g^{\prime}\right)\right)(c)\right) \\ & =-\ln \left(\max \left\{w_{1}^{\prime} f^{\prime} c, w^{\prime} 2 g^{\prime} c\right\}\right) \\ & =\min \left\{-\ln \left(w_{1}^{\prime} f^{\prime} c\right),-\ln \left(w_{2}^{\prime} g^{\prime} c\right)\right\} \\ & =\min \left\{-\ln \left(f^{\prime} c\right)-\ln \left(w_{1}^{\prime}\right),-\ln \left(g^{\prime} c \right)-\ln \left(w_{2}^{\prime}\right)\right\} \\ & = \min \left\{f c+w_{1}, g c+w_{2}\right\} . \end{aligned}$	$\begin{aligned} ((w_{1}, f) \sqcup (w_{2}, g))(c) & =-\ln \left(((w_{1}^{\prime}, f^{\prime}) \sqcup (w_{2}^{\prime}, g^{\prime}))(c)\right) \\ & =-\ln \left(\max \left\{w_{1}^{\prime} f^{\prime} c, w^{\prime} 2 g^{\prime} c\right\}\right) \\ & =\min \left\{-\ln \left(w_{1}^{\prime} f^{\prime} c\right),-\ln \left(w_{2}^{\prime} g^{\prime} c\right)\right\} \\ & =\min \left\{-\ln(f^{\prime} c)-\ln(w_{1}^{\prime}),-\ln(g^{\prime} c)-\ln(w_{2}^{\prime})\right\} \\ & = \min \left\{f c+w_{1}, g c+w_{2}\right\} . \end{aligned}$
2106.07890_EQ0050	024	■ 1.000 ● N|+0	`s_{1} \oplus s_{2}=\min \left\{s_{1}, s_{2}\right\} \text { and } s_{1} \odot s_{2}=s_{1}+s_{2}`	$s_{1} \oplus s_{2}=\min \left\{s_{1}, s_{2}\right\} \text { and } s_{1} \odot s_{2}=s_{1}+s_{2}$	$s_{1} \oplus s_{2}=\min\{s_{1}, s_{2}\} \text { and } s_{1} \odot s_{2}=s_{1}+s_{2}$
2106.07890_EQ0051	025	■ 1.000 ● N|+0	`\left(\left(w_{1}, f\right) \sqcup\left(w_{2}, g\right) \right)(c)=\min \left\{f c+w_{1}, g c+w_{2}\right\}=w_{1} \odot f c \oplus w_{2} \odot g c .`	$\left(\left(w_{1}, f\right) \sqcup\left(w_{2}, g\right) \right)(c)=\min \left\{f c+w_{1}, g c+w_{2}\right\}=w_{1} \odot f c \oplus w_{2} \odot g c .$	$((w_{1}, f) \sqcup (w_{2}, g))(c)=\min \{f c+w_{1}, g c+w_{2}\}=w_{1} \odot f c \oplus w_{2} \odot g c.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Image regions — rendered against scan

Two comparable cells per row. **Rendered**: the region's own LaTeX compiled as its OWN document, or — where no LaTeX exists — the scan again, so every row has two cells and the ink difference reads as a floor rather than as missing-versus-present. A duplicated row is marked *(dup)* and its distance is a SELF-comparison, not agreement between two sources. **Scan**: the `tex.zip` image whose filename region 5-tuple matches this row, else the CDN crop. The Class column carries the residual class beside MathPix's confidence, as the equation rows do. Setting a region's LaTeX does NOT say it renders to what is on the page — that is what the two columns are for, and 281 is the open question they exist to let a reader answer. A region with no LaTeX can be reconstructed with `pdfdrill vision`; verify any LLM result against the real ink with inkdrill.

Identifier	Page	Class	Source	Author source	Rendered	Scan
2106.07890_DIA_0001	005	— *(dup)*	88b8a351-0ce2-4826-bc11-f6b9d764d111-05_344_1036_397_546.jpg			
2106.07890_DIA_0002	009	— *(dup)*	88b8a351-0ce2-4826-bc11-f6b9d764d111-09_344_1038_397_544.jpg			
2106.07890_DIA_0003	012	— *(dup)*	88b8a351-0ce2-4826-bc11-f6b9d764d111-12_176_348_597_889.jpg			
2106.07890_DIA_0004	016	— *(dup)*	88b8a351-0ce2-4826-bc11-f6b9d764d111-16_226_223_859_952.jpg			
2106.07890_DIA_0005	016	— *(dup)*	88b8a351-0ce2-4826-bc11-f6b9d764d111-16_292_438_1302_842.jpg			
2106.07890_DIA_0006	022	— *(dup)*	88b8a351-0ce2-4826-bc11-f6b9d764d111-22_339_1063_395_531.jpg